Dungeoneers: Investigating Cybersickness Acclimatization in Virtual Reality

CS 465, Colorado State University

Angel Gonzalez Portillo
Joshua Masih
John Robert Harrison Matthews

Our Problem Statement: the Start

- Cybersickness remains a major barrier to VR adoption.
- Symptoms include nausea, vertigo, eye strain, and disorientation.
- New users are especially vulnerable.

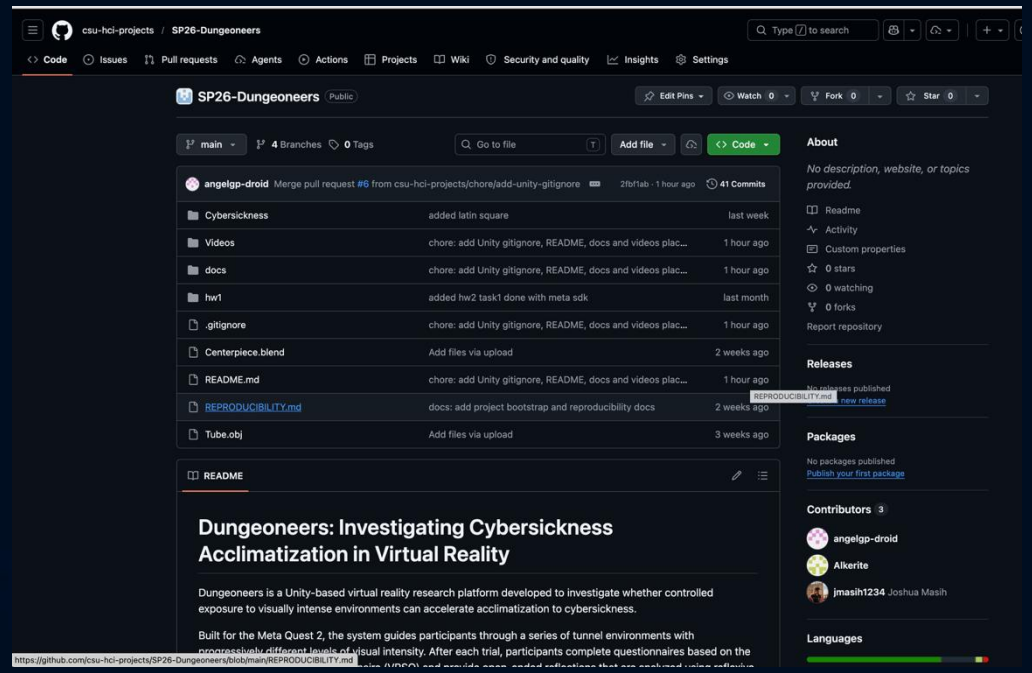
Research Question: Does prior exposure to a high-intensity VR environment accelerate acclimatization to lower-intensity environments?

Related Work

- Thomas A. Stoffregen and Joseph J. LaViola Jr.: Sensory conflict theory
- Paul M. Gavgani: CARE (Cybersickness Abatement from Repeated Exposure)
- Stephen R. Palmisano: Adaptation can be stimulus-specific
- Our study builds on Sensory Conflict Theory and CARE research, uses the VRSQ developed by Kim et al. to measure symptoms, and applies Braun and Clarke's thematic analysis framework to interpret qualitative responses.

Related Work System Overview: Dungeoneers

- Continuous locomotion
- Voice commands (“unlock” / “lock”)
- Wrist-based dial rotation
- Automatic CSV logging



Related Work Methodology:

- Methodology
- Study Design
- 6 participants
- Within-subjects
- Latin square counterbalancing
- 20–30 minute sessions
- VRSQ questionnaire after each trial
- Semi-structured interviews
- Participant Breakdown
- Experienced, moderate, novice, and first-time VR users

Interaction Flow

Step 1

- Press B to start
- Walk through tunnel

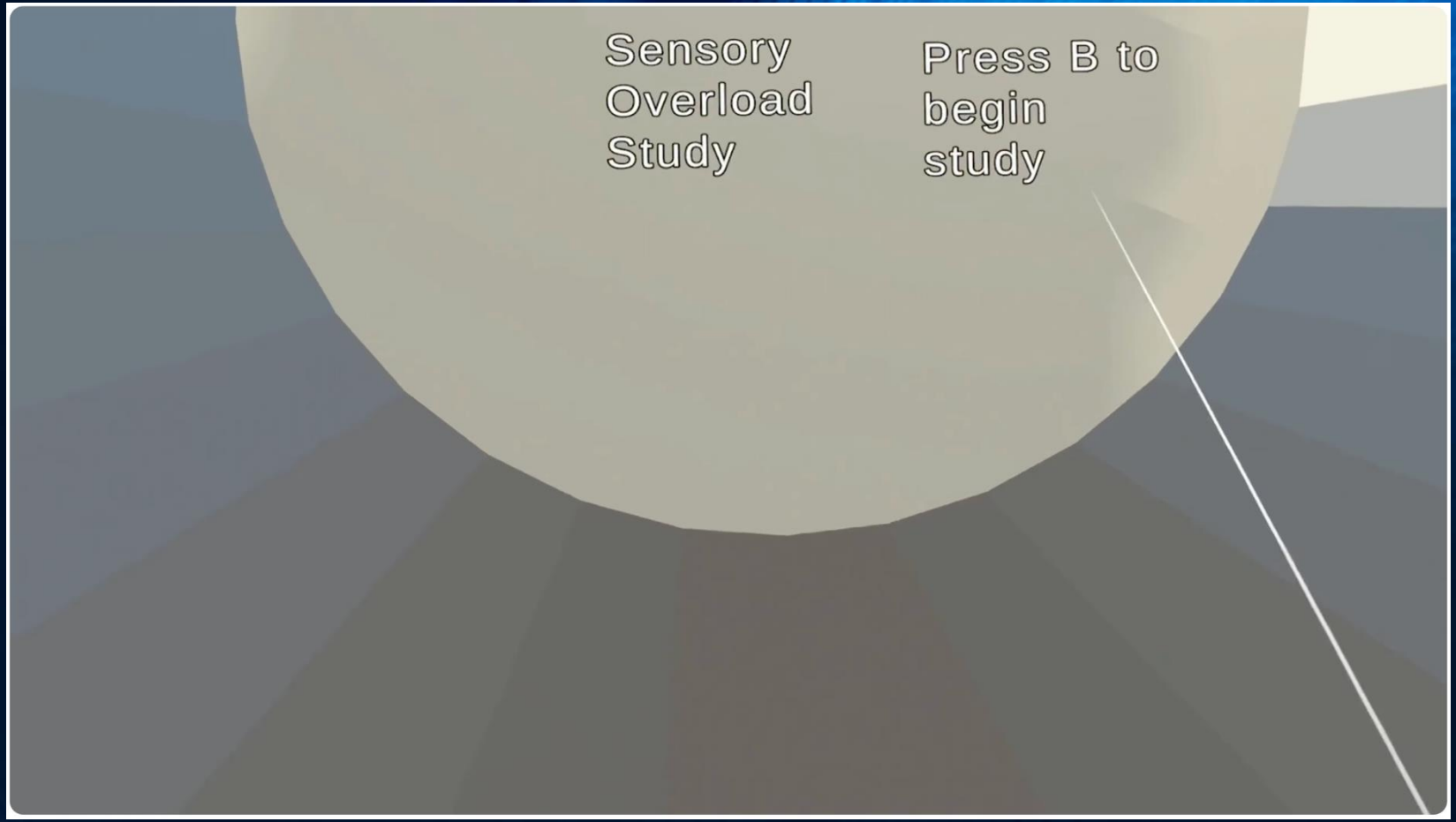
Step 2

- Say “Unlock”
- Rotate dial 180°

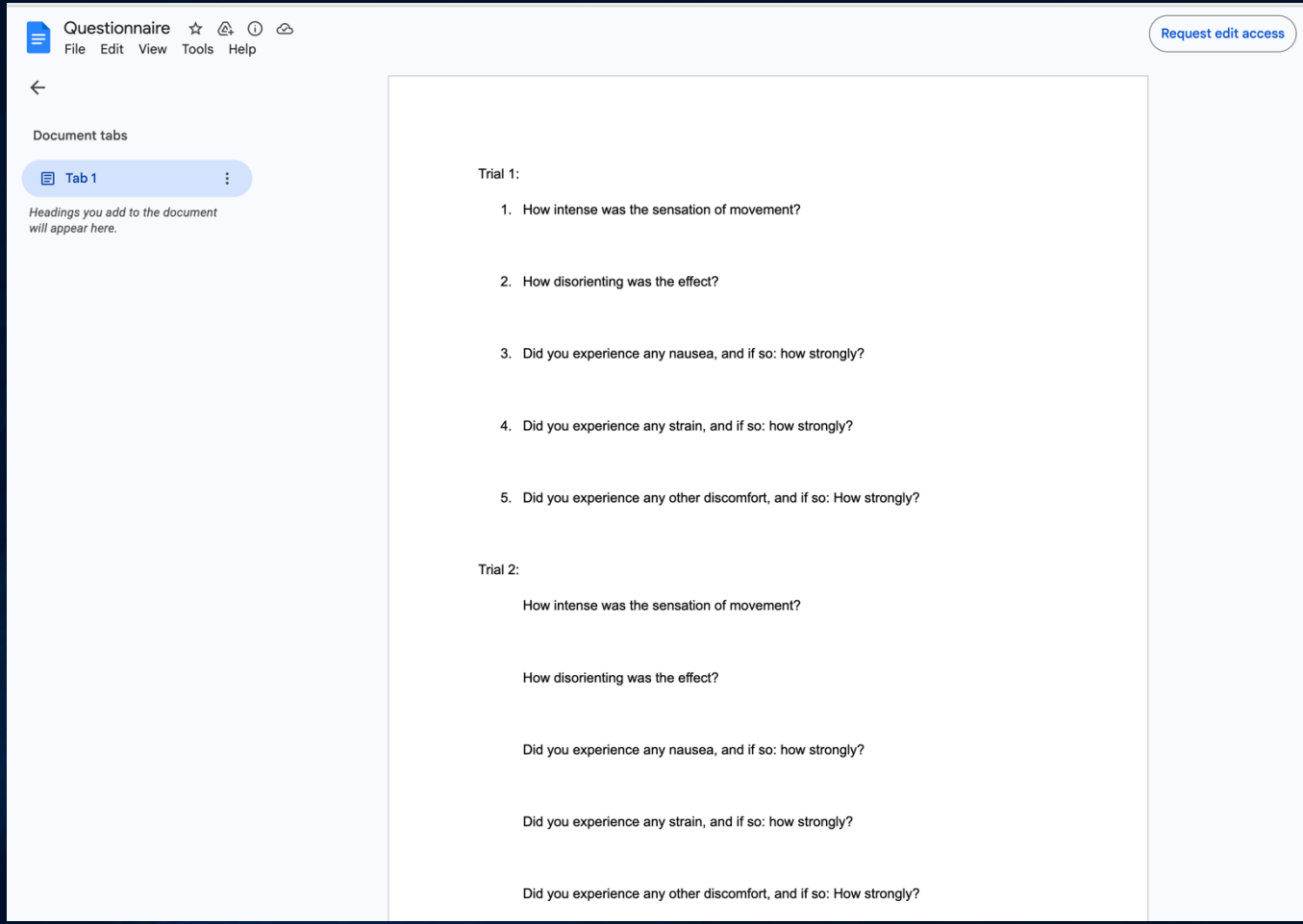
Step 3

- Say “Lock”
- Complete questionnaire

Visuals for Procedure: Control Condition



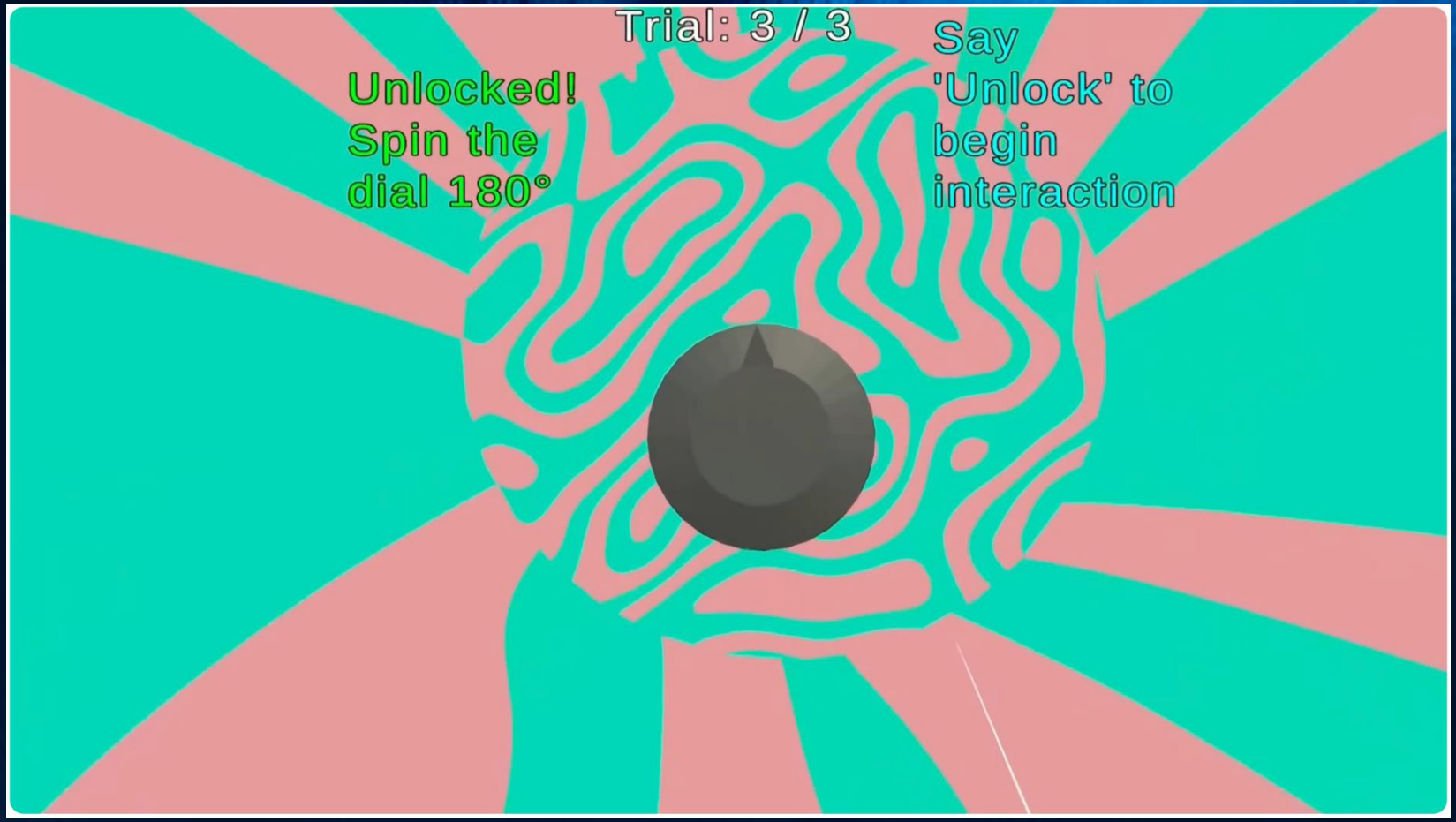
In-between Trials



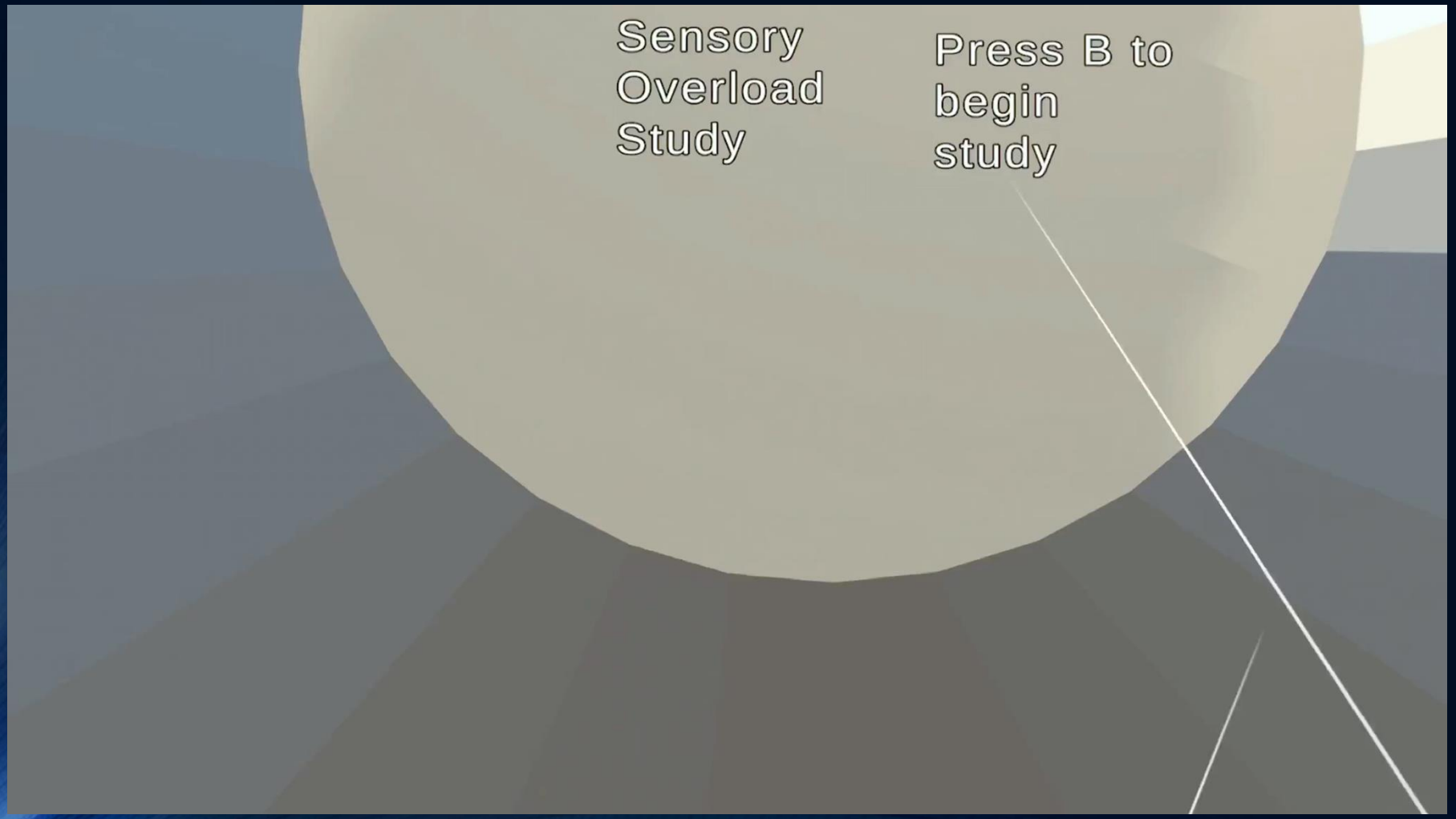
Visuals for Procedure: CONDITION 1 Purple and Yellow (high intensity)

Visuals for Procedure: CONDITION 2 black and white (medium intensity)

Visuals for Procedure: CONDITION 3 Pink and Blue (low intensity)



Demo



Results

- Purple/Yellow produced the strongest vertigo, eye strain, and nausea.
- Black/White also induced notable discomfort.
- Pink/Blue was consistently described as easier to navigate.
- Control produced almost no discomfort.
- Most participants became more comfortable over successive trials.
- First-time VR users experienced the strongest lingering symptoms.

Future Work

- Recruit more participants
- Add physiological measures (heart rate, GSR)
- Conduct longitudinal studies
- Develop adaptive real-time intensity adjustment

Thematic Analysis

MAJOR THEMES IDENTIFIED

1. Experience Shapes Cybersickness

- Experienced VR users reported minimal discomfort.
- Novice and first-time users experienced stronger symptoms (vertigo, nausea, eye strain).
- Prior exposure to VR appears to improve tolerance.

2. Cybersickness ≠ Motion Sickness

- History of car or airplane sickness did not reliably predict VR discomfort.
- Suggests cybersickness is a distinct phenomenon driven by sensory conflict.

3. Trial Order and Acclimatization

- Purple/yellow condition was the most provocative.
- Later conditions often felt easier after high-intensity exposure.
- Evidence of short-term adaptation within a single session.

Conclusion

What We Learned:

•Controlled exposure to visually intense environments may accelerate short-term acclimatization.

•Cybersickness is influenced by both:

- Environmental design
- User experience level

•High-contrast moving patterns reliably provoke symptoms.

•Repeated exposure appears to reduce discomfort during a single session.